

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 1.2 — Managing Billing Configuration

GCP Associate Cloud Engineer

Study Notes Study Notes April 2026

Exam Relevance

This topic is part of **Section 1: Setting up a cloud solution environment (~20 % of the exam)**. You must know how to create billing accounts, link projects to them, establish budgets and alerts, and set up billing exports.

1. Google Cloud Billing Overview

How Billing Works

- **Every project must be linked to a billing account** to create billable resources
- Charges are accumulated at the project level and rolled up to the billing account
- A billing account can be linked to one or more projects
- Payment is processed at the billing account level

Billing Account Types

Type	Description
Self-serve (online)	Linked to a credit/debit card or bank account. Costs charged automatically.
Invoiced (offline)	Available to eligible customers. Google sends monthly invoices. Requires application and approval.

Billing Account vs. Project Relationship

```
Billing Account
├─ Project A (charges: $150/month)
├─ Project B (charges: $300/month)
└─ Project C (charges: $50/month)
    Total invoice: $500/month
```

2. Creating Billing Accounts

Permissions Required

- **Billing Account Creator** (`roles/billing.creator`) — at the organization level to create new billing accounts
- **Billing Account Administrator** (`roles/billing.admin`) — manage existing billing accounts

Steps to Create a Billing Account

1. Go to the **Cloud Console** → **Billing**
2. Click **Manage Billing Accounts** → **Create Account**
3. Enter account name, country, and currency
4. Select payment method (credit card, bank account, or invoiced)
5. Complete verification

gcloud Commands

```
# List all billing accounts accessible to you
gcloud billing accounts list

# Describe a billing account
gcloud billing accounts describe BILLING_ACCOUNT_ID

# List projects linked to a billing account
gcloud billing projects list --billing-account=BILLING_ACCOUNT_ID
```

Key Points

- A billing account can be linked to **multiple projects**
- A project can only be linked to **one billing account** at a time
- Billing accounts are separate from the resource hierarchy (they sit alongside it)

- Organization-level billing accounts can be managed centrally

3. Linking Projects to a Billing Account

Why Link Projects?

- Without a billing account, a project cannot create resources that incur charges
- Free-tier resources may still work, but billable resources will fail to create

How to Link

```
# Link a project to a billing account
gcloud billing projects link PROJECT_ID \
  --billing-account=BILLING_ACCOUNT_ID

# Unlink a project from billing (disables billing — resources may be stopped)
gcloud billing projects unlink PROJECT_ID

# Check which billing account a project is linked to
gcloud billing projects describe PROJECT_ID
```

Permissions Required

- **Billing Account User** (`roles/billing.user`) on the billing account AND
- **Project Owner** or **Project Billing Manager** on the project

What Happens When You Unlink Billing?

- **Resources are not immediately deleted**, but:
 - Compute Engine VMs are stopped
 - Cloud Storage buckets become read-only
 - Most services stop functioning
 - After a grace period, resources may be deleted
 - Re-linking restores billing but stopped resources must be manually restarted
-

4. Establishing Billing Budgets and Alerts

Billing Budgets

A budget lets you track your actual (and forecasted) Google Cloud spending against a planned amount.

Types of Budget Amounts

- **Specified amount** — A fixed dollar amount (e.g., \$1,000/month)
- **Last month's spend** — Dynamically set based on the previous month's charges

Budget Scope

You can scope a budget to: - Entire billing account - Specific projects - Specific products/services - Specific labels

Alert Thresholds

- Default thresholds: **50 %**, **90 %**, **100 %** of budget
- You can customize these percentages
- Alerts can be based on **actual spend** or **forecasted spend**

Alert Notification Options

1. **Email to billing admins and users** — Default notification method
2. **Cloud Monitoring notification channels** — Email, SMS, Slack, PagerDuty, etc.
3. **Pub/Sub topic** — For programmatic responses (e.g., auto-disable billing, send custom alerts)

Setting Up Budgets via Console

1. Go to **Billing → Budgets & Alerts**
2. Click **Create Budget**
3. Name the budget and select scope (billing account, projects, services)
4. Set the budget amount
5. Configure alert thresholds
6. Choose notification method

Programmatic Budget Response (Pub/Sub)

Budget Alert → Pub/Sub Topic → Cloud Function → Action

└─ Disable billing

└─ Scale down resources

└─ Send Slack notification

└─ Cap spending

Key Exam Points

- **Budgets do NOT cap spending** — They only send notifications
- To actually stop spending, you need a programmatic solution (Cloud Function + Pub/Sub)
- Forecasted alerts can warn you before you hit the threshold
- Budgets can be scoped to projects, services, or labels for granular tracking

5. Setting Up Billing Exports

Billing exports allow you to send detailed billing data to external destinations for analysis and reporting.

Export Destinations

Destination	Use Case	Detail Level
BigQuery	Detailed analysis, custom reports, dashboards	Line-item detail (most granular)
Cloud Storage (CSV/JSON)	Archival, external tool ingestion	Daily summary

BigQuery Export (Recommended)

Types of BigQuery Export

1. **Standard usage cost** — Detailed cost and usage data (most commonly used)
2. **Detailed usage cost** — Even more granular (resource-level) cost data
3. **Pricing data** — Export of Google Cloud pricing information

Setting Up BigQuery Export

1. Go to **Billing → Billing Export**

2. Select the **BigQuery Export** tab
3. Choose or create a BigQuery dataset
4. Select the export type(s)
5. Click **Save**

Key Properties

- Export is **near real-time** (data appears within hours)
- Historical data is NOT backfilled (export starts from the enablement date)
- The dataset must be in the same organization as the billing account
- BigQuery export is **free** (you only pay for queries on the exported data)

Example BigQuery Query

```
-- Total cost by service for the current month
SELECT
  service.description AS service,
  SUM(cost) AS total_cost,
  SUM(IFNULL((SELECT SUM(c.amount) FROM UNNEST(credits) c), 0)) AS total_credits
FROM
  `project.dataset.gcp_billing_export_v1_XXXXXX`
WHERE
  invoice.month = FORMAT_DATE('%Y%m', CURRENT_DATE())
GROUP BY
  service
ORDER BY
  total_cost DESC;
```

```
-- Daily cost trend
SELECT
  DATE(usage_start_time) AS usage_date,
  SUM(cost) AS daily_cost
FROM
  `project.dataset.gcp_billing_export_v1_XXXXXX`
WHERE
  invoice.month = FORMAT_DATE('%Y%m', CURRENT_DATE())
GROUP BY
  usage_date
ORDER BY
  usage_date;
```

Cloud Storage Export (Legacy)

- Exports daily CSV or JSON files to a Cloud Storage bucket
- Less detailed than BigQuery export

- Useful for feeding data into external tools
- **Note:** Google recommends BigQuery export as the primary method

6. Billing Roles and Permissions

Key Billing IAM Roles

Role	Description
<code>roles/billing.creator</code>	Create new billing accounts
<code>roles/billing.admin</code>	Manage billing accounts (view costs, link/unlink projects, manage payments)
<code>roles/billing.user</code>	Link projects to billing accounts
<code>roles/billing.viewer</code>	View billing account cost and transactions
<code>roles/billing.projectManager</code>	Link/unlink projects from a billing account
<code>roles/billing.costsManager</code>	Manage budgets and view/export costs (no payment management)

Best Practices for Billing Roles

- Use **billing.viewer** for teams that need cost visibility but no management rights
- Use **billing.user** for project creators who need to link new projects to billing
- Use **billing.admin** sparingly — it has broad permissions including payment management
- Assign billing roles at the **organization level** for centralized control

7. Cost Management Tools

Pricing Calculator

- Estimate costs before deploying resources
- Available at cloud.google.com/products/calculator
- Model different configurations and compare costs

Cost Table (Billing Reports)

- Visual breakdown of costs in the Cloud Console
- Filter by project, service, SKU, time period, and labels
- View trends, forecasts, and cost anomalies

Committed Use Discounts (CUDs)

- 1-year or 3-year commitments for Compute Engine and Cloud SQL
- **Savings:** Up to 57% for 1-year, up to 70% for 3-year
- Applies to CPU and memory (not GPUs or premium OS licenses)
- Purchased at the project level

Sustained Use Discounts (SUDs)

- **Automatic discounts** for running Compute Engine resources more than 25% of the month
- No commitment required
- Up to 30% discount for full-month usage
- Applied automatically at the project level

Labels for Cost Allocation

- Key-value pairs attached to resources (e.g., `environment:production`, `team:backend`)
- Used to filter and group costs in billing reports
- Can be used as budget scope criteria

```
# Add labels to a VM instance
gcloud compute instances add-labels INSTANCE_NAME \
  --labels=environment=production,team=backend

# Add labels to a Cloud Storage bucket
gcloud storage buckets update gs://BUCKET_NAME \
  --update-labels=environment=staging
```

8. Cost Controls Beyond Budgets

Budgets alert but do **NOT** stop spending — they are notification-only. To actually stop spending when a budget is exceeded, combine budgets with programmatic controls:

Option 1 — Disable billing via Pub/Sub + Cloud Function:


```
# Cloud Function triggered by budget Pub/Sub notification
def stop_billing(data, context):
    import base64, json
    from googleapiclient import discovery
    pubsub_data = json.loads(base64.b64decode(data['data']).decode('utf-8'))
    if pubsub_data['costAmount'] >= pubsub_data['budgetAmount']:
        # Remove billing account from project
        billing = discovery.build('cloudbilling', 'v1')
        billing.projects().updateBillingInfo(
            name='projects/MY_PROJECT',
            body={'billingAccountName': ''}
        ).execute()
```

Option 2 — Organization policy to restrict resources (e.g., deny VM creation in certain regions) — manual but governance-based.

Option 3 — Quotas: request a quota reduction to cap resource usage (e.g., limit number of VMs or CPUs per region).

Exam tip: A common exam scenario asks how to PREVENT overspending — the answer is NOT budgets alone; it's budgets + Pub/Sub + Cloud Function to disable billing, or quota reductions.

9. Billing Export and Analysis

```
# Enable billing export to BigQuery
# (Must be done via Console: Billing → Billing export → BigQuery export → Edit settings)
# Then query in BigQuery:
SELECT
    service.description AS service,
    SUM(cost) AS total_cost,
    SUM(IFNULL((SELECT SUM(c.amount) FROM UNNEST(credits) c), 0)) AS total_credits,
    SUM(cost) + SUM(IFNULL((SELECT SUM(c.amount) FROM UNNEST(credits) c), 0)) AS net_cost
FROM `PROJECT.DATASET.gcp_billing_export_v1_ACCOUNT_ID`
WHERE DATE(usage_start_time) BETWEEN DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY) AND CURRENT_DATE()
GROUP BY 1 ORDER BY net_cost DESC;

-- Cost by project
SELECT project.name, SUM(cost) AS total FROM `...` GROUP BY 1 ORDER BY 2 DESC;

-- Cost by label (e.g., team=backend)
SELECT labels.value AS team, SUM(cost)
FROM `...`, UNNEST(labels) AS labels
WHERE labels.key = 'team'
GROUP BY 1;
```

- **Standard export:** daily granularity, resource-level cost
- **Detailed export (recommended):** hourly granularity, VM/disk-level cost, includes resource hierarchy
- Exam tip: Billing export to BigQuery is the only way to do fine-grained historical cost analysis; Cloud Console billing shows real-time but limited drill-down

10. Billing Account Hierarchy

```
Organization
├── Billing Account(s)
│   └── Linked Projects
│       └── Resources (charges roll up to project, then billing account)
```

Key Exam Points on Hierarchy

- A **billing account** exists at the **organization level** (not within folders/projects)
- A billing account can be linked to projects in different folders
- Multiple billing accounts can exist under one organization (e.g., per department)
- Billing account admins are different from project admins

Exam Practice Questions

1. **A project creator needs to set up a new project and link it to the company's billing account. Which roles do they need?**
 - Answer: `roles/billing.user` on the billing account AND `roles/resourcemanager.projectCreator` on the organization or folder.
2. **Your team wants to automatically shut down resources when spending exceeds \$5,000. How do you accomplish this?**
 - Answer: Create a budget with a Pub/Sub notification. Create a Cloud Function triggered by the Pub/Sub topic that disables billing on the project (or scales down resources).
3. **You need to analyze your Google Cloud spending by team and environment. What should you do?**
 - Answer: Apply labels (team, environment) to all resources. Enable BigQuery billing export. Query the export data filtered by labels.

4. What happens to running VMs if a project's billing account is removed?

- Answer: The VMs will be stopped. They won't be immediately deleted, but they won't run. After a grace period, the project and its resources may be deleted.

5. Which billing export method provides the most detailed cost data?

- Answer: BigQuery export (specifically, the "Detailed usage cost" export type) provides the most granular, resource-level cost data.

6. Can a billing budget automatically stop spending when the threshold is reached?

- Answer: No. Billing budgets only send alerts/notifications. To cap spending, you need a programmatic solution using Pub/Sub and Cloud Functions.

Glossary

AES-256 — Advanced Encryption Standard with a 256-bit key; the encryption algorithm used by GCP to protect data at rest by default.

Billing Account — A GCP resource that defines who is responsible for paying for a set of linked projects; exists at the organization level and processes payments.

Billing Account Administrator ([roles/billing.admin](#)) — IAM role granting full management of a billing account, including viewing costs, linking/unlinking projects, and managing payments.

Billing Account Creator ([roles/billing.creator](#)) — IAM role granted at the organization level that permits creation of new billing accounts.

Billing Account User ([roles/billing.user](#)) — IAM role that allows linking projects to a billing account; typically combined with Project Owner or Project Billing Manager.

Billing Account Viewer ([roles/billing.viewer](#)) — IAM role granting read-only access to billing account cost data and transactions.

Billing Budget — A GCP construct that tracks actual and forecasted spending against a defined amount and sends alert notifications when thresholds are crossed; does not cap spending.

Billing Export — Feature that sends detailed billing data to BigQuery or Cloud Storage for analysis and reporting purposes.

Billing Project Manager ([roles/billing.projectManager](#)) — IAM role allowing the holder to link or unlink projects from a billing account without full billing admin rights.

BigQuery — Google Cloud's serverless data warehouse; the recommended destination for billing exports due to its fine-grained, queryable, near-real-time cost data.

BYOL (Bring Your Own License) — Licensing model where a customer uses their existing software licenses on cloud infrastructure, relevant for Sole-Tenant Nodes on Compute Engine.

Cloud Console — The web-based graphical interface for managing GCP resources; used for creating billing accounts, setting budgets, and configuring billing exports.

Cloud Function — Serverless, event-driven function used in billing cost-control scenarios to programmatically disable billing when a Pub/Sub budget alert is triggered.

Cloud SQL — GCP's managed relational database; eligible for Committed Use Discounts alongside Compute Engine.

Cloud Storage — GCP's object storage service; a destination for legacy billing exports (CSV/JSON) and a resource whose costs appear in billing reports.

Coldline — Cloud Storage class designed for data accessed less than once per quarter; lower storage cost but higher retrieval cost than Nearline.

Committed Use Discounts (CUDs) — Discounts of up to 57% (1-year) or 70% (3-year) on Compute Engine CPU and memory in exchange for a usage commitment; purchased at the project level.

Compute Engine — GCP's virtual machine service; subject to Sustained Use Discounts and Committed Use Discounts.

Cost Anomaly — An unexpected spike or change in spending visible in the Cloud Console billing reports and cost table.

Costs Manager ([roles/billing.costsManager](#)) — IAM role allowing management of budgets and viewing/exporting costs without payment management capabilities.

CSV (Comma-Separated Values) — File format used by the legacy Cloud Storage billing export method.

Dataflow — GCP's managed stream and batch data processing service; can appear as a line item in billing reports.

Detailed Usage Cost Export — The most granular BigQuery billing export type, providing hourly, resource-level (VM/disk) cost data including resource hierarchy information.

Firestore — GCP's serverless NoSQL document database; appears as a billable service line item.

Free Tier — A set of GCP resources available at no cost each month, including limited Compute Engine, Cloud Storage, BigQuery, and other service usage.

GCP (Google Cloud Platform) — Google's suite of cloud computing services.

gcloud CLI — Command-line interface for GCP; used to list billing accounts, link/unlink projects, and query project billing status.

GPU (Graphics Processing Unit) — Specialized hardware accelerator; excluded from Committed Use Discounts on Compute Engine.

IAM (Identity and Access Management) — GCP's access control system; used to assign billing-specific roles to control who can manage billing accounts and budgets.

Invoiced Billing — A billing account type available to eligible large customers where Google sends monthly invoices instead of charging a card automatically.

JSON (JavaScript Object Notation) — File format used by the legacy Cloud Storage billing export method alongside CSV.

Label — A key-value pair (e.g., `environment:production`) attached to GCP resources; used to filter and group costs in billing reports and as budget scope criteria.

Nearline — Cloud Storage class for data accessed less than once per month; lower storage cost but higher retrieval cost than Standard.

OLAP (Online Analytical Processing) — Workload category focused on complex analytical queries over large datasets; BigQuery is designed for OLAP.

OLTP (Online Transactional Processing) — Workload category focused on fast read/write transactions; Cloud SQL and Spanner serve OLTP workloads.

Organization — Top-level node in the GCP resource hierarchy; billing accounts exist at this level and can be linked to projects across any folder.

PagerDuty — Third-party incident management platform that can receive Cloud Monitoring alert notifications triggered by billing budget thresholds.

Pricing Calculator — Google Cloud tool at cloud.google.com/products/calculator for estimating GCP resource costs before deployment.

Project — Core GCP organizational unit; charges accumulate at the project level and roll up to the linked billing account.

Project Billing Manager — See `roles/billing.projectManager`; a role allowing project-to-billing-account linking without full billing admin permissions.

Pub/Sub (Cloud Pub/Sub) — GCP's asynchronous messaging service; used as the notification channel for billing budget alerts to enable programmatic spending controls.

Self-Serve Billing — A billing account type linked to a credit/debit card or bank account where charges are processed automatically; the default for most GCP customers.

SKU (Stock Keeping Unit) — A unique identifier for a specific GCP resource or service configuration used to itemize costs in billing reports.

Slack — Third-party collaboration tool that can receive Cloud Monitoring alert notifications, including billing budget alerts.

Spanner (Cloud Spanner) — GCP's globally distributed relational database; appears as a billable service line item.

Standard Usage Cost Export — BigQuery billing export type providing daily, resource-level cost and usage data; the most commonly used export type.

Sustained Use Discounts (SUDs) — Automatic Compute Engine discounts of up to 30% applied when a VM runs for more than 25% of a billing month; requires no commitment.

VPC (Virtual Private Cloud) — GCP network resource whose egress traffic can incur billing charges, visible in billing reports.